

Table 6: Simulation VI: Empirical power and Monte Carlo standard error for $\pi = 0.10$, under different values of μ and r .

$r = 2$									
n	$\mu = 0.5$			$\mu = 1.0$			$\mu = 1.5$		
	Uniform	Central	Tail	Uniform	Central	Tail	Uniform	Central	Tail
50	0.0000 (0.0000)	0.0470 (0.0030)	0.0520 (0.0031)	0.0000 (0.0000)	0.0354 (0.0026)	0.0642 (0.0035)	0.0000 (0.0000)	0.0344 (0.0026)	0.0930 (0.0041)
100	0.0000 (0.0000)	0.0510 (0.0031)	0.0562 (0.0033)	0.0000 (0.0000)	0.0432 (0.0029)	0.0562 (0.0033)	0.0000 (0.0000)	0.0278 (0.0023)	0.0802 (0.0038)
200	0.0000 (0.0000)	0.0546 (0.0032)	0.0526 (0.0032)	0.0000 (0.0000)	0.0426 (0.0029)	0.0554 (0.0032)	0.0000 (0.0000)	0.0330 (0.0025)	0.0752 (0.0037)
500	0.0000 (0.0000)	0.0546 (0.0032)	0.0474 (0.0030)	0.0000 (0.0000)	0.0582 (0.0033)	0.0486 (0.0030)	0.0000 (0.0000)	0.0234 (0.0021)	0.0898 (0.0040)
1000	0.0000 (0.0000)	0.0468 (0.0030)	0.0466 (0.0030)	0.0000 (0.0000)	0.0478 (0.0030)	0.0496 (0.0031)	0.0000 (0.0000)	0.0194 (0.0020)	0.1064 (0.0044)

$r = 5$									
n	$\mu = 0.5$			$\mu = 1.0$			$\mu = 1.5$		
	Uniform	Central	Tail	Uniform	Central	Tail	Uniform	Central	Tail
50	0.0000 (0.0000)	0.0402 (0.0028)	0.0478 (0.0030)	0.0000 (0.0000)	0.0358 (0.0026)	0.0696 (0.0036)	0.0000 (0.0000)	0.0286 (0.0024)	0.1298 (0.0048)
100	0.0000 (0.0000)	0.0460 (0.0030)	0.0536 (0.0032)	0.0000 (0.0000)	0.0372 (0.0027)	0.0832 (0.0039)	0.0000 (0.0000)	0.0228 (0.0021)	0.1448 (0.0050)
200	0.0000 (0.0000)	0.0462 (0.0030)	0.0510 (0.0031)	0.0000 (0.0000)	0.0342 (0.0026)	0.0802 (0.0038)	0.0000 (0.0000)	0.0170 (0.0018)	0.1566 (0.0051)
500	0.0000 (0.0000)	0.0466 (0.0030)	0.0518 (0.0031)	0.0000 (0.0000)	0.0330 (0.0025)	0.0902 (0.0041)	0.0000 (0.0000)	0.0088 (0.0013)	0.2382 (0.0060)
1000	0.0000 (0.0000)	0.0470 (0.0030)	0.0548 (0.0032)	0.0000 (0.0000)	0.0166 (0.0018)	0.1168 (0.0045)	0.0000 (0.0000)	0.0024 (0.0007)	0.4004 (0.0069)

$r = 8$									
n	$\mu = 0.5$			$\mu = 1.0$			$\mu = 1.5$		
	Uniform	Central	Tail	Uniform	Central	Tail	Uniform	Central	Tail
50	0.0000 (0.0000)	0.0496 (0.0031)	0.0580 (0.0033)	0.0000 (0.0000)	0.0412 (0.0028)	0.0920 (0.0041)	0.0000 (0.0000)	0.0326 (0.0025)	0.1402 (0.0049)
100	0.0000 (0.0000)	0.0410 (0.0028)	0.0694 (0.0036)	0.0000 (0.0000)	0.0356 (0.0026)	0.1100 (0.0044)	0.0000 (0.0000)	0.0236 (0.0021)	0.1870 (0.0055)
200	0.0000 (0.0000)	0.0396 (0.0028)	0.0560 (0.0033)	0.0000 (0.0000)	0.0262 (0.0023)	0.1042 (0.0043)	0.0000 (0.0000)	0.0146 (0.0017)	0.2188 (0.0058)
500	0.0000 (0.0000)	0.0442 (0.0029)	0.0628 (0.0034)	0.0000 (0.0000)	0.0248 (0.0022)	0.1382 (0.0049)	0.0000 (0.0000)	0.0030 (0.0008)	0.3756 (0.0068)
1000	0.0000 (0.0000)	0.0362 (0.0026)	0.0694 (0.0036)	0.0000 (0.0000)	0.0114 (0.0015)	0.1814 (0.0054)	0.0000 (0.0000)	0.0006 (0.0003)	0.6068 (0.0069)

Table 7: Simulation VI: Empirical power and Monte Carlo standard error for $\pi = 0.20$, under different values of μ and r .

$r = 2$									
n	$\mu = 0.5$			$\mu = 1.0$			$\mu = 1.5$		
	Uniform	Central	Tail	Uniform	Central	Tail	Uniform	Central	Tail
50	0.0000 (0.0000)	0.0412 (0.0028)	0.0634 (0.0034)	0.0000 (0.0000)	0.0328 (0.0025)	0.0938 (0.0041)	0.0000 (0.0000)	0.0210 (0.0020)	0.1600 (0.0052)
100	0.0000 (0.0000)	0.0436 (0.0029)	0.0574 (0.0033)	0.0000 (0.0000)	0.0288 (0.0024)	0.0896 (0.0040)	0.0000 (0.0000)	0.0154 (0.0017)	0.1604 (0.0052)
200	0.0000 (0.0000)	0.0490 (0.0031)	0.0610 (0.0034)	0.0000 (0.0000)	0.0310 (0.0025)	0.0908 (0.0041)	0.0000 (0.0000)	0.0120 (0.0015)	0.1892 (0.0055)
500	0.0000 (0.0000)	0.0464 (0.0030)	0.0630 (0.0034)	0.0000 (0.0000)	0.0236 (0.0021)	0.1236 (0.0047)	0.0000 (0.0000)	0.0050 (0.0010)	0.2958 (0.0065)
1000	0.0000 (0.0000)	0.0396 (0.0028)	0.0694 (0.0036)	0.0000 (0.0000)	0.0118 (0.0015)	0.1522 (0.0051)	0.0000 (0.0000)	0.0010 (0.0004)	0.4502 (0.0070)

$r = 5$									
n	$\mu = 0.5$			$\mu = 1.0$			$\mu = 1.5$		
	Uniform	Central	Tail	Uniform	Central	Tail	Uniform	Central	Tail
50	0.0000 (0.0000)	0.0328 (0.0025)	0.0730 (0.0037)	0.0000 (0.0000)	0.0254 (0.0022)	0.1380 (0.0049)	0.0000 (0.0000)	0.0114 (0.0015)	0.2986 (0.0065)
100	0.0000 (0.0000)	0.0316 (0.0025)	0.0696 (0.0036)	0.0000 (0.0000)	0.0182 (0.0019)	0.1616 (0.0052)	0.0000 (0.0000)	0.0050 (0.0010)	0.3616 (0.0068)
200	0.0000 (0.0000)	0.0318 (0.0025)	0.0738 (0.0037)	0.0000 (0.0000)	0.0138 (0.0016)	0.2132 (0.0058)	0.0000 (0.0000)	0.0024 (0.0007)	0.4910 (0.0071)
500	0.0000 (0.0000)	0.0330 (0.0025)	0.0828 (0.0039)	0.0000 (0.0000)	0.0034 (0.0008)	0.3162 (0.0066)	0.0000 (0.0000)	0.0000 (0.0000)	0.7670 (0.0060)
1000	0.0000 (0.0000)	0.0160 (0.0018)	0.1160 (0.0045)	0.0000 (0.0000)	0.0010 (0.0004)	0.4992 (0.0071)	0.0000 (0.0000)	0.0000 (0.0000)	0.9564 (0.0029)

$r = 8$									
n	$\mu = 0.5$			$\mu = 1.0$			$\mu = 1.5$		
	Uniform	Central	Tail	Uniform	Central	Tail	Uniform	Central	Tail
50	0.0000 (0.0000)	0.0508 (0.0031)	0.0718 (0.0037)	0.0000 (0.0000)	0.0248 (0.0022)	0.1608 (0.0052)	0.0000 (0.0000)	0.0150 (0.0017)	0.3290 (0.0066)
100	0.0000 (0.0000)	0.0398 (0.0028)	0.0898 (0.0040)	0.0000 (0.0000)	0.0150 (0.0017)	0.2304 (0.0060)	0.0000 (0.0000)	0.0044 (0.0009)	0.4736 (0.0071)
200	0.0000 (0.0000)	0.0276 (0.0023)	0.0848 (0.0039)	0.0000 (0.0000)	0.0068 (0.0012)	0.2792 (0.0063)	0.0000 (0.0000)	0.0002 (0.0002)	0.6286 (0.0068)
500	0.0000 (0.0000)	0.0272 (0.0023)	0.1132 (0.0045)	0.0000 (0.0000)	0.0016 (0.0006)	0.4708 (0.0071)	0.0000 (0.0000)	0.0000 (0.0000)	0.9074 (0.0041)
1000	0.0000 (0.0000)	0.0158 (0.0018)	0.1756 (0.0054)	0.0000 (0.0000)	0.0004 (0.0003)	0.7004 (0.0065)	0.0000 (0.0000)	0.0000 (0.0000)	0.9938 (0.0011)

Table 8: Simulation VI: Empirical power and Monte Carlo standard error for $\pi = 0.30$, under different values of μ and r .

$r = 2$									
n	$\mu = 0.5$			$\mu = 1.0$			$\mu = 1.5$		
	Uniform	Central	Tail	Uniform	Central	Tail	Uniform	Central	Tail
50	0.0000 (0.0000)	0.0378 (0.0027)	0.0716 (0.0036)	0.0000 (0.0000)	0.0250 (0.0022)	0.1242 (0.0047)	0.0000 (0.0000)	0.0078 (0.0012)	0.2440 (0.0061)
100	0.0000 (0.0000)	0.0384 (0.0027)	0.0760 (0.0037)	0.0000 (0.0000)	0.0182 (0.0019)	0.1416 (0.0049)	0.0000 (0.0000)	0.0090 (0.0013)	0.2694 (0.0063)
200	0.0000 (0.0000)	0.0366 (0.0027)	0.0804 (0.0038)	0.0000 (0.0000)	0.0166 (0.0018)	0.1626 (0.0052)	0.0000 (0.0000)	0.0032 (0.0008)	0.3644 (0.0068)
500	0.0000 (0.0000)	0.0322 (0.0025)	0.0966 (0.0042)	0.0000 (0.0000)	0.0076 (0.0012)	0.2592 (0.0062)	0.0000 (0.0000)	0.0002 (0.0002)	0.5704 (0.0070)
1000	0.0000 (0.0000)	0.0208 (0.0020)	0.1202 (0.0046)	0.0000 (0.0000)	0.0024 (0.0007)	0.4178 (0.0070)	0.0000 (0.0000)	0.0000 (0.0000)	0.8088 (0.0056)

$r = 5$									
n	$\mu = 0.5$			$\mu = 1.0$			$\mu = 1.5$		
	Uniform	Central	Tail	Uniform	Central	Tail	Uniform	Central	Tail
50	0.0000 (0.0000)	0.0336 (0.0025)	0.0852 (0.0039)	0.0000 (0.0000)	0.0184 (0.0019)	0.2210 (0.0059)	0.0000 (0.0000)	0.0024 (0.0007)	0.4614 (0.0070)
100	0.0000 (0.0000)	0.0290 (0.0024)	0.0906 (0.0041)	0.0000 (0.0000)	0.0072 (0.0012)	0.2788 (0.0063)	0.0000 (0.0000)	0.0010 (0.0004)	0.5906 (0.0070)
200	0.0000 (0.0000)	0.0250 (0.0022)	0.1124 (0.0045)	0.0000 (0.0000)	0.0028 (0.0007)	0.3860 (0.0069)	0.0000 (0.0000)	0.0002 (0.0002)	0.7570 (0.0061)
500	0.0000 (0.0000)	0.0140 (0.0017)	0.1700 (0.0053)	0.0000 (0.0000)	0.0002 (0.0002)	0.6436 (0.0068)	0.0000 (0.0000)	0.0000 (0.0000)	0.9576 (0.0028)
1000	0.0000 (0.0000)	0.0064 (0.0011)	0.2674 (0.0063)	0.0000 (0.0000)	0.0000 (0.0000)	0.8884 (0.0045)	0.0000 (0.0000)	0.0000 (0.0000)	0.9988 (0.0005)

$r = 8$									
n	$\mu = 0.5$			$\mu = 1.0$			$\mu = 1.5$		
	Uniform	Central	Tail	Uniform	Central	Tail	Uniform	Central	Tail
50	0.0000 (0.0000)	0.0366 (0.0027)	0.0988 (0.0042)	0.0000 (0.0000)	0.0150 (0.0017)	0.2640 (0.0062)	0.0000 (0.0000)	0.0046 (0.0010)	0.5308 (0.0071)
100	0.0000 (0.0000)	0.0272 (0.0023)	0.1320 (0.0048)	0.0000 (0.0000)	0.0086 (0.0013)	0.3790 (0.0069)	0.0000 (0.0000)	0.0000 (0.0000)	0.7124 (0.0064)
200	0.0000 (0.0000)	0.0182 (0.0019)	0.1472 (0.0050)	0.0000 (0.0000)	0.0016 (0.0006)	0.5208 (0.0071)	0.0000 (0.0000)	0.0000 (0.0000)	0.8596 (0.0049)
500	0.0000 (0.0000)	0.0094 (0.0014)	0.2430 (0.0061)	0.0000 (0.0000)	0.0000 (0.0000)	0.8060 (0.0056)	0.0000 (0.0000)	0.0000 (0.0000)	0.9916 (0.0013)
1000	0.0000 (0.0000)	0.0030 (0.0008)	0.3586 (0.0068)	0.0000 (0.0000)	0.0000 (0.0000)	0.9682 (0.0025)	0.0000 (0.0000)	0.0000 (0.0000)	1.0000 (0.0000)